EGB220 Mechatronics Design I
Supplied Parts
Version 1 - 2026

Each Team will be provided with <u>one</u> EGB220 Line and Light Following Kit. The kit will be signed out by a team member using their Student ID in week 1. The kits contain components for the practicals and to start developing and testing your prototypes. Refer to the EGB220 Challenge Rules document on Canvas for details on which components of the kit can be used in the final product without penalty.

Figure 1 shows a kit and what must be returned to the S9 store at the end of the unit.

Item	To be returned to S9 store at end of the Unit
Line Follower Kit Plastic case	Yes
Spark	Yes
SparkBOB	Yes
Sensor Array	Yes
EGB220 Payload BOB	Yes
XT30 Battery Charger	Yes
XT30 7.4V Li-Ion Battery	Yes
Micro USB cable	Yes
Teensy & Mini USB cable	Yes
2 x sensor PCB's	No
2 x Infrared emitters	No
2 x Infrared phototransistors	No
2 x Servo motors	Yes
2 x 50:1 micro gear motors	Yes
Breadboard	Yes
Box of jumper wires	Yes

Figure 1: The EGB220 Line Following Kit contents with what must be returned at the end of the unit.

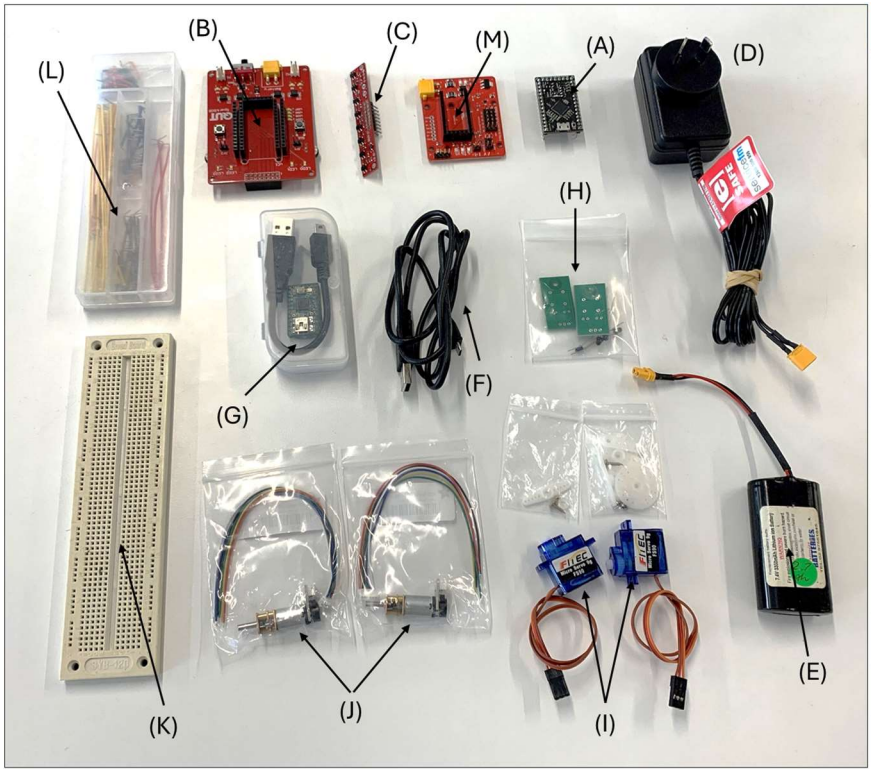
Figure 2 highlights each item in the kit with the following outlining some details of the hardware, and if used in your final product, the dollar amount which needs to be included in your product budget.

Motors

The two supplied motors (Figure 2) are 50:1 Micro Gear Motors with magnetic encoders and a cable to assist with wiring. These were purchased in bulk and can be costed at **$15.00 each**. Please do not cut encoder wires off the connector (even if not using). You will find details about the motors here: https://www.adafruit.com/product/4638

Servos

The two supplied servos (Figure 2) are 9g Micro Servo FS90 (180 degree, 1.5kg/cm) and cost **$5.70 each**. You will find details about them here: https://core-electronics.com.au/feetech-fs90-1-5kgcm-micro-servo-9g.html



Item
(A) Spark
(B) SparkBOB
(C) Reflectance Sensor Array
(D) XT30 Battery Charger
(E) XT30 7.5V Li-Ion Battery
(F) Micro USB Cable
(G) Teensy & Mini USB cable
(H) 2 x sensor PCBs, 2 x Infrared Transmitters (LEDs), 2 x Infrared phototransistors
(I) 2 x Servo motors
(J) 2 x 50:1 micro gear motors (with plastic servo horns)
(K) Breadboard
(L) Box of jumpers wires
(M) EGB220 Payload BOB

Figure 2: The supplied EGB220 Line Following Kit contents.

Microcontroller (Spark)

The **Spark** microprocessor board (Figure 2) was created by QUT's EESS and can be costed in your budget at the discounted price of **$15**. Most teams will use this in the final demo and there is no penalty for using it (other than cost). The schematic for the Spark is on Blackboard.

Spark Break Out Board (SparkBOB)

The **SparkBOB** (Figure 2) has been created by the S9 technicians (it is a breakout board that you can use in the early days of development) and is ***free for you to use during development***. If you do use it in your final product, it will cost you **$20**. But most teams will design and build their own BOB as you will score more highly for that. The schematic for the SparkBOB is on Blackboard.

Payload Break Out Board (Payload BOB)

The EGB220 **Payload BOB** (Figure 2) has been created by the teaching team to interface with the Teensy microcontroller for early-stage development of the IR light tracking system. It is ***free for you to use during development***, however, if you do use it in your final product, it will cost you **$20**. But most teams will design and build their own Payload BOB for the Teensy as you will score more highly for that. The schematic for the Payload BOB is on Blackboard.

ATmega 32u4 microcontroller module (Teensy)

A Teensy 2.0 (Figure 2) is supplied. This has the same microcontroller as the SparkBOB (an ATmega 32u4) but without motor controllers and battery power management. This has been supplied for development and use for the IR light tracking payload. It will plug directly onto the EGB220 Payload BOB. If you use it in your final product, it will cost you **$20**. Most teams will use this in the Final Demo and there is no penalty for using it (other than cost).

Sensor Board

The sensor board (**QTR-8A Reflectance Sensor Array)** that we hand out will cost **$16.95** if you include it in your final product. Details of that PCB can be found here:
https://www.robotgear.com.au/Product.aspx/Details/3543-QTR-8A-Reflectance-Sensor-Array

Again, we expect that most teams will make their own sensor board and use their own board in the Final Demo as you score more highly for that.

Battery

The battery available from the S9 store is a 2-cell Lithium Ion battery, giving out 7.4V with a capacity of 3400mAh. It has internal protection against over voltage, under voltage, over current and short circuit. For the purposes of your robot budget, the **battery costs $5**.

These batteries **must be returned** to the S9 store at the end of the unit and will be collected at the end of the Final Demo.

Battery Charger

The battery charger is provided for you to use but does not need to be counted in the budget. However, like the battery, it also **must be returned** to the S9 store at the end of the unit.

Breadboard

You will use the breadboard in Prac 2 (Week 2) and might want to use it as you prototype your circuits.

IR LEDs and Phototransistors

2 x Infrared emitter : TSAL4400 (worth $0.45 each)
https://au.element14.com/vishay/tsal4400/ir-emitter-3mm-940nm/dp/1045420

2 x Infrared phototransistor : TEFT4300 (worth $0.63 each)
https://au.element14.com/vishay/teft4300/phototransistor-t1/dp/1045526?ost=1045526&ddkey=https%3Aen-AU%2FElement14_Australia%2Fsearch

You will use these in Practical 1 (Week 2).

USB Micro Cable

This is used to connect your laptop or PC to the Spark and will need to be returned to the S9 store with the rest of your supplied kit at the end of the unit.

USB Mini Cable

This is used to connect your laptop or PC to the Teensy 2.0 and will need to be returned to the S9 store with the rest of your supplied kit at the end of the unit.

Additional (not in kit) Hardware

During the Practical sessions, a selection of wheels, casters, microswitches, LEDs, and motor brackets will be made available for teams to use in your prototype if they wish. These are optional to use in your product design and reduces the need for individual ordering of common parts and wait times. However, any of these parts are used, they must be costed at current market rates from Core Electronics or Digikey (in Australian dollars) for your product budget.